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Executive Summary of the Thesis

# Asset Management Techniques under ESG constraints: Efficient Frontier Analysis and Portfolio Weights Forecasting with a Kalman Filter

Laurea Magistrale in Mathematical Engineering - Ingegneria Matematica

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## Introduction

The integration of Environmental, Social and Governance (ESG) principles into investment processes has become a cornerstone of modern asset management. The emergence of regulatory frameworks, together with a growing awareness of climate change and corporate responsibility, has progressively aligned capital allocation with sustainability objectives.

Asset managers must reconcile ethical coherence with financial efficiency, ensuring that portfolios remain profitable while satisfying sustainability criteria. The classical Markowitz mean variance model still provides the foundation for quantitative portfolio construction, but it was not originally designed to incorporate non-financial objectives. This raises a central question: how do explicit ESG constraints reshape the efficient frontier?

The thesis addresses this question by integrating two methodological layers.

The first is a set of static portfolio optimization techniques under ESG constraints, implemented through alternative approaches: ESG thresholds, negative screening, linear limits on low-rated assets and Best-in-Class integration.

The second is a dynamic modeling framework based on the Kalman Filter (KF), designed to forecast the Minimum Variance Portfolio weights implemented in each section of the first chapter, in order to have an Out Of Sample evaluation of the models.

The empirical analysis focuses on S&P 500 constituents from 2015 to 2024, using monthly data for returns and ESG Scores from Refinitiv.

Portfolios are evaluated under both long-only and limited short-selling regimes, using Annualized Return (AR), Volatility (Vol), Sharpe Ratio (SR), Maximum Drawdown (MDD) and Calmar Ratio (CR) as performance metrics.

## 1. Asset management techniques considering ESG scores

Asset management aims to balance risk and return through the efficient allocation of capital across diversified assets. The mean–variance model minimizes portfolio variance for a given expected return, providing a systematic method to construct efficient frontiers. Extreme values are winsorized at the 5% tails to reduce estimation noise and remove outliers. Two allocation regimes are tested: long-only

(non-negative weights) and limited short-selling (weights bounded below by -10% of portfolio composition).
Moreover, in every section a qualitative analysis of the frontier is done evaluating the plots and a quantitative one is computed evaluating the performance of the Minimum Variance Portfolio.

### 1.1. Refinitiv ESG Scores

ESG scores quantify firms' sustainability performance along environmental, social and governance dimensions. They have become essential to investment decision-making, influencing capital flows and corporate policies. Integrating ESG factors can occur through rigid constraints, exclusions, soft linear limits or relative ranking. The design of these mechanisms defines the trade-off between ethical alignment and portfolio efficiency.
ESG Scores are taken from Refinitiv: they go from 0 to 100 and they are then normalized removing the mean of the ESG Scores of the assets in the same way as Barucci, Azzone and Stocco [1].

### 1.2. Literature Review

Azzone, Barucci and Stocco (2025)[1] propose a formal framework for ESG-constrained optimization, showing that efficiency losses remain limited when sustainability targets are properly calibrated. In the thesis, it is adopted a mean–variance setup, not Tracking Error Variance based, and the ESG constraint is considered as a portfolio-level score constraint.
Blitz and Swinkels (2019, 2023)[3] document that broad exclusion policies reduce diversification, while selective and well-calibrated screening preserves performance. In this thesis, the negative screening approach is applied at the sector level, excluding industries characterized by structurally low ESG standards to evaluate the aggregate effect of sectoral bans on portfolio efficiency.
De Spiegeleer et al. (2020)[4] demonstrate that sector-aware Best-in-Class integration can align sustainability goals with diversification, keeping the efficient frontier close to the baseline. The reference is thus used to justify the sector-aware construction and the interpretation of Best-in-Class, which forces short selling for assets with low ESG score.
Together, these works define the theoretical foundation for the ESG integration methods and performance analysis developed in this thesis.

### 1.3. Data Filtration

The dataset includes S&P 500 constituents with price data from 2015 to 2019 and ESG scores as of 2019. To ensure a manageable and representative universe, only the top 100 companies by market capitalization are retained, since most of the 500 assets would otherwise remain inactive in the efficient frontier construction.
After filtering for liquidity and missing data, 93 assets remain.

### 1.4. Standard and Top ESG Score Frontiers Analysis

The unconstrained frontier is used as a theoretical introduction for all the frontiers presented in the thesis.

$$\begin{aligned} \min_{w \in R^n} \quad & w^\top \Sigma w \\ \text{s.t.} \quad & \mathbf{1}^\top w = 1, \\ & \mu^\top w = \mu_p \\ & w_i \geq 0 \quad \forall i. \end{aligned}$$

Allowing mild short-selling, so changing the constraint with $w_i \geq -0.1 \quad \forall i$, increases performance and diversification, but it is not stable from a computational point of view.
Focusing only on different Top-ESG subset, so evaluating the frontier considering {10, 19, 28, ..., 93} assets with highest ESG Scores, the frontiers with less assets are dominated by the more populated ones because, removing assets, diversification decreases.
Moreover, this first analysis suggests a possible negative premium related to increasing ESG constraint.
From this base, two investable universes are formed: the Standard universe of 93 assets, representing broad market exposure, and the Top-ESG subset of 50 firms with the highest sustainability ratings. This dual setup allows the study of concentration and quality effects on portfolio efficiency.

### 1.5. ESG-Constrained Optimization

Adding a minimum portfolio-level ESG threshold introduces a measurable cost

$$\sum_{i=1}^{n} w_i \, \mathrm{ESG}_i \geq G$$

with G going from 0 to the maximum ESG score among assets in the optimization framework.
Increasing the value of G, the frontiers are dominated by the ones with softer constraints. Moreover, volatility increases and Sharpe ratios decreases: in the full universe, volatility rises from 8% to 13% as the ESG threshold moves from neutral to stringent, while SR declines from 1.5 to 1.1.
Allowing small short positions partially offsets this effect, improving Sharpe and Calmar Ratio.
In the Top-ESG universe, constraints bind earlier and costs rise faster, but moderate thresholds still yield balanced trade-offs.

### 1.6. Negative Screening - Sectorial Analysis

Negative screening removes entire sectors with low ESG ratings: in this section, {1, 2, ..., 10} sectors are removed in the same way as the Top ESG Frontiers analysis.
Moderate exclusions, up to five sectors, may improve risk-adjusted returns, but excessive bans severely limit diversification. Results confirm the non-linear impact of exclusions: moderate screening enhances SR, which reach 1.4, while over-screening increases risk concentration, indeed Maximum Drawdown reaches also -20% and volatility is higher than other portfolios.
In this case short selling does not improve results, but the performance is mainly caused by the reduction of the available assets, which reduces diversification.

### 1.7. ESG Integration via Linear Limits on Low-Score Assets

Linear limits provide a softer approach, capping the aggregate exposure to low-rated assets.

$$\sum_{i \in \mathcal{L}} w_i \quad \leq \gamma$$

Limiting the worst {10, 20, 30, 40} ESG stocks, which correspond to the set $\mathcal{L}$ in the disequation, to a percentage of total weight, in this case $\gamma = 0.1$, maintains diversification while improving sustainability: in the full universe soft exclusions bring good results in terms of returns, around 13-14%, and Maximum Drawdown.
In the Top-50 universe, such limits produce stable performance with Volatility around 8–10%, SR near 1.3, and CR near 1.8. Allowing limited short-selling further enhances returns and efficiency, with MDD smaller than -5% and SR close to 1.8. The cases with few limited assets perform slightly better, highlighting that flexible integration delivers better outcomes than rigid exclusion.
This is the only case where the Top 50 ESG Score performs slightly better than the full one. The problem is related to the cap constraint because weights are more concentrated and induce unstable frontiers that are statistically and economically less reliable.

### 1.8. Best-in-Class Portfolios

In the Best-in-Class approach, the assets with smaller ESG Scores are forced to be only short sold: this is a consequence of what it is noticed before, since it is empirically observed a negative premium for assets with low ESG Score.
While Best-in-Class with long-only weights for the best ESG rated assets do not perform better than other models, the case with allowed short selling achieves the strongest performance among all static designs, combining high ESG exposure with robust efficiency.
In the full universe with short selling allowed, SR averages 1.5–1.6 and CR exceeds 2, with drawdowns and volatility below 10%. In particular, this is observed in the case with 10-20 assets with forced short selling: this confirms again that soft constraints bring good performance.
These results confirm that sector-aware ESG integration outperforms hard thresholds.

### 1.9. Overall Results

In this first chapter, it is observed that stricter constraints are associated with a negative ESG premium, as tighter sustainability requirements systematically reduce portfolio efficiency. Portfolios restricted to the Top-50 ESG scores underperform relative to broader universes; therefore, this subset is excluded from the dynamic analysis presented in the next chapter.
On the other hand, portfolios with soft con-

straints in the Best-in-Class and Constrained Optimization cases get good results, especially in terms of volatility reduction and Sharpe Ratio performance.
Moreover, allowing limited short-selling improves frontier performance and diversification, but introduces numerical instability and, usually, higher volatility.

## 2. Weights Forecasting

The second chapter extends the static portfolio analysis toward a dynamic dimension, introducing a predictive framework designed to evaluate the temporal evolution of portfolio weights. The objective is not to forecast returns directly, but to model how optimal allocations evolve through time, capturing persistence and gradual structural shifts in Minimum Variance Portfolio Weights composition.
The Kalman Filter (KF) is employed as a state-space model that regularizes and smooths portfolio weights across time, producing consistent and continuous trajectories even in the presence of noisy market inputs.
The analysis focuses on forecasting the minimum-variance portfolio: data from 2015–2022 are first used to construct the set of portfolios required to initialize and estimate the Kalman Filter. Then, for each month of 2023, the corresponding portfolio is forecasted, and its performance is evaluated out of sample over the following 12 months, including 2024.

### 2.1. Kalman Filter Overview

The Kalman Filter (KF) is an optimal recursive estimation technique for systems that evolve linearly over time and are subject to Gaussian noise. It decomposes the dynamics of portfolio weights into two components: a latent state transition and a noisy observation process:

$$x_t = Ax_{t-1} + w_t, \quad w_t \sim \mathcal{N}(0, Q),$$

$$y_t = Hx_t + v_t, \quad v_t \sim \mathcal{N}(0, R),$$

where $x_t$ denotes the latent portfolio weights, $y_t$ represents the observed weights, $A$ and $H$ are transition and observation matrices and $Q$ and $R$ are the process and observation noise covariances. In practice, $y_t$ corresponds to the estimated weights and the state dynamics are estimated for each asset using the historical sequence of Minimum Variance Weights stored in $W$. For each asset $i$, an AR(1) model is fitted to capture the persistence of portfolio weights:

$$w_{i,t} = \phi_i \, w_{i,t-1} + \eta_{i,t}, \qquad \eta_{i,t} \sim \mathcal{N}(0, q_i).$$

The persistence coefficient $\phi_i$ is estimated as

$$\phi_i = \frac{\sum_t w_{i,t-1} w_{i,t}}{\sum_t w_{i,t-1}^2},$$

which is clipped between [0, 0.999] and it represents the strength of the linear dependence between consecutive weights. The process variance $q_i$ is computed as the variance of the residuals $e_{i,t} = w_{i,t} - \phi_i w_{i,t-1}$. These parameters are then arranged into diagonal matrices

$$A = \mathrm{diag}(\phi_1, \ldots, \phi_n), \qquad Q = \mathrm{diag}(q_1, \ldots, q_n).$$

The measurement model assumes $H = I_n$ and a small constant noise $R = \sigma_\varepsilon^2 I_n$, with $\sigma_\varepsilon^2 = 10^{-4}$. The initial state $x_0$ is set to the last observed weight vector, and each forecasted vector is clipped and normalized to ensure portfolio feasibility.
The KF captures inertia in the allocation process: weight changes occur gradually unless significant new information generates a strong innovation in the Kalman gain.

#### Metrics

To evaluate the quality of forecasts, predicted weights are compared to realized (optimized) ones using standard error and performance metrics, including the $L_1$ and $L_2$ distances, Cosine Similarity, return-based $L_1^{\text{return}}$ and $L_2^{\text{return}}$, as well as Annualized Return, Annualized Volatility, Sharpe Ratio, Maximum Drawdown and Calmar Ratio which evaluate KF performance.
The Kalman-based forecast does not aim to maximize short-term returns, but to produce stable, continuous weight trajectories that retain economic meaning and operational coherence.

#### Limitations and Scope

A key limitation of the Kalman Filter framework concerns short-selling configurations. When short positions are allowed, the predicted weights often become unstable and not economically relevant, exhibiting excessive volatility and erratic fluctuations. The short-selling case, despite achieving competitive returns in the first

chapter analysis, lacks computational stability and economical interpretability; therefore, it is not included in the forecasting evaluation.

### 2.2. ESG Integration via Best-in-Class: Forecasting Results

Quantitatively, performance metrics show that the predictive efficiency of the model remains solid throughout the 12-month horizon. Annualized returns and volatility are comparable to those of the realized portfolios, with differences increasing slightly in the final months due to the cumulative effect of the one-step-ahead propagation. The Sharpe ratio obtained with KF remains stable around 2-2.5 with 10 assets with forced short selling and decreases limiting more assets. The Calmar ratio follows a similar pattern, it stays around 6-8, reflecting consistent performance under controlled risk.
At the structural level, the forecast alignment weakens slightly over time: the average $L_1$ distance between forecasted and realized weights increases progressively. $L_2$ distance shows the same tendency. Nevertheless, the cosine similarity remains high, around 0.8–0.9.
The graphical comparison supports these findings, showing trajectories that diverge gradually but remain smooth and stable over time. Despite this, the overall dynamics remain coherent and no abrupt shifts are observed, especially in cases with 10-20 assets with limited investments.

### 2.3. ESG Integration via Linear Limits on Low-Score Assets: Forecasting Results

Performance metrics show that the predictive efficiency of the model remains robust throughout the 12-month horizon. Annualized returns remain close to those of the realized portfolios, while volatility decreases moderately in the first months and rises slightly toward the end of the forecast window due to accumulated uncertainty. The Sharpe ratio stays again within a range 2-2.5. In this case, Volatility is slightly slower than the Best-in-Class case, here it is less than 15% in most situations, but also the returns slightly decreases.
As before, the average $L_1$ distance between predicted and realized weights increases across months, and the $L_2$ distance exhibits a similar trend. Cosine similarity remains high.
In the plot analysis, these observations are confirmed and overall the Kalman Filter effectively models the persistence of portfolio weights under linear ESG constraints.

### 2.4. ESG Integration via Constrained Optimization: Forecasting Results

Quantitatively, performance metrics show that the predictive efficiency of the model remains solid across all constraint levels. Annualized returns stay close to those of the realized portfolios, while volatility increases moderately as ESG bounds tighten—reflecting the reduced diversification typical of constrained universes. The Sharpe ratio remains stable, again close to 2-2.5, in the early forecast periods and slightly decreases for the strictest thresholds, indicating that the Kalman Filter maintains acceptable risk-adjusted performance even under higher sustainability targets.
Moreover, MDD are smaller than -5% underlying lower risks.
At the structural level, the forecast alignment weakens slightly as constraints become more binding: with respect to other cases, in this part metrics shows very good results even considering the last months of the analysis: Cosine similarity is usually greater than 0.9.
The graphical comparison confirms these findings, showing that trajectories for both low and high ESG thresholds remain smooth, though deviations grow as the constraint tightens.

### 2.5. Negative Screening – Sectorial Analysis: Forecasting Results

Performance metrics show that the model maintains good predictive consistency at moderate exclusion levels. Annualized returns stay close to those of the realized portfolios, while volatility gradually rises with the number of excluded sectors, reflecting a decline in diversification. Thae Sharpe ratio remains stable in the early forecast periods, around 1.2–1.6, and decreases slightly in the final months.
Calmar Ratio is too low, usually way smaller than 5, and, in general, performance are very different from the real situation: this means that from an economical point of view this portfolio composition is not reliable.
Forecast alignment weakens progressively: both

the $L_1$ and $L_2$ distances between predicted and realized weights increase with the exclusion intensity, while cosine similarity in some cases reaches also values close to zero.
Overall, the Kalman Filter proves that dynamic smoothing improves weight continuity and coherence with soft exclusions. On the other hand, performance is worse than all other portfolios, especially volatility which is higher than 20-25% in many cases.

### 2.6. Overall Discussion and Interpretation

The Kalman Filter (KF) proves to be an effective dynamic forecasting tool for portfolio weights, offering smoother and more stable allocations than static optimizations. Across all ESG configurations, it delivers high annualized returns and Sharpe ratios close to real portfolios, but with lower volatility and smaller drawdowns — particularly under moderate ESG constraints and Best-in-Class or Linear-Limit setups. Its adaptive nature enhances stability and robustness, capturing market dynamics without overreacting to short-term noise. Compared to real portfolios, the KF maintains high predictive accuracy and risk-adjusted performance, confirming its ability to reproduce efficient frontiers while mitigating volatility, especially when diversification remains sufficient. Overall, the model successfully adds a dynamic and resilient dimension to ESG-constrained portfolio management.

## 3. Discussion and Comparison with the Literature

### 3.1. Summary of Findings

The analysis shows that ESG integration reshapes portfolio efficiency without implying systematic underperformance. Hard or exclusionary constraints reduce diversification and increase volatility, while softer methods, in particular in linear constraints cases and Best-in-Class selection, maintain stable risk–return profiles. Dynamic forecasting with the Kalman Filter further enhances stability, smoothing weight trajectories and preserving efficiency even under ESG restrictions.

### 3.2. Interpretation of the Empirical Results

From an economic standpoint, the evidence points to a persistent negative ESG premium: portfolios with stricter sustainability constraints systematically underperform their unconstrained counterparts in risk–adjusted terms. This inefficiency arises because high-ESG assets, often perceived as safer or more mature, tend to offer lower expected returns, while exclusionary rules reduce exposure to risk premia embedded in low-ESG sectors. The resulting trade-off is structural: stronger ESG alignment comes at the cost of reduced financial efficiency.

### 3.3. Alignment with the Literature

The results are consistent with previous research. Azzone et al. (2025)[1] confirm that ESG thresholds reduce performance, while Blitz and Swinkels (2023)[3] find that moderate exclusions have limited performance impact. De Spiegeleer (2020)[4] supports the effectiveness of Best-in-Class strategies in preserving diversification.

### 3.4. Future Developments

Future work could extend this framework by incorporating climate-related metrics such as carbon intensity and net-zero alignment, following the approaches of Bolton et al. (2022)[6] and Roncalli (2022)[8]. Additional research directions include the use of robust optimization to handle ESG score uncertainty, the inclusion of transaction costs, and the exploration of nonlinear or hybrid filtering techniques to improve adaptability. Finally, extending the Kalman-based forecasting to portfolios with controlled short-selling exposure represents a key step toward a more comprehensive and realistic dynamic model.

### 3.5. Concluding Remarks

The thesis shows that responsible investing can remain both efficient and practical when supported by adaptive, data-driven methods. Integrating ESG scores through soft constraints or limiting particular assets, rather than implementing rigid optimization constraints, offers a balanced framework that successfully aligns sustainability goals with solid financial performance.

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